

CHATGPT - VOICE INPUT

Say the word, start the work—enterprise voice that's fast, accurate, and private by design.

TEAM: PRODUCT

STATUS: BRAINSTORMING

CONTRIBUTOR : DIVYA

RESOURCE: MILESTONE 1 &2

Problem definition

- **Problem:**
 - Voice input is hard to find, lacks of Chatgpt assistance “hey chatgpt”, and is unreliable—so users default to “Hey Google”.
 - While text input remains stable, the voice experience often suffers from mid-session disruptions during long prompts due to **network latency** and **buffer timeouts**, necessitating a transition to **Incremental Chunking** and **Local Audio Caching** to match the reliability of our text-based processing.
- **Who:** Employees and interview-prep users needing fast, hands-free input.
- **Business value:** More engagement and retention; less reliance on third-party assistants that may pose security/compliance risks.
- **User benefit:** Quicker hands-free access, higher accuracy, and stable long-form dictation.
- **Urgency:** Competitors are shaping habits now; fixing this prevents churn and keeps data in approved channels.

Goals

- Enable seamless, reliable voice-first access (including optional wake word) to increase adoption.
- Improve transcription accuracy and stability for long/complex prompts.
- Boost awareness and onboarding for voice features within the app.
- Ensure enterprise-grade security, privacy, and compliance for voice data.

Non - goals

- No OS-level/global hotword or third-party assistant integrations (Google/Siri/Alexa).
- No broad multilingual expansion or offline/on-device ASR this release (cloud-backed, limited languages only).
- No advanced analytics/coaching features or custom training on proprietary audio beyond transient, policy-aligned processing.

Measurable metrics (with targets)

- **Voice session share:** +30% relative increase within 90 days.
- **Chatgpt assistance success rate:** ≥97% across supported devices.
- **ASR word error rate (WER):** ≤7% on benchmark datasets; ≤10% in-field.
- **Streaming stability:** <1% mid-session drop rate; P95 latency to first token <800 ms.

- **Feature discoverability:** ≥60% of active users engage voice at least once per month.
- **Task success rate (voice-only flows):** ≥85% (measured via completion without retry).

Why these metrics matter

- Adoption and discoverability indicate product-market fit for voice workflows.
- Accuracy, latency, and stability directly drive trust and repeat usage.
- Task success and CSAT reflect real user value, not just usage volume.

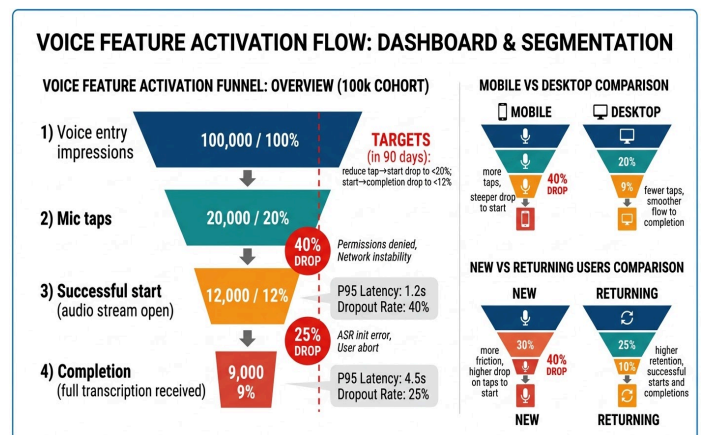
Validation summary

User research/data

- Discovery gap: 62% of surveyed employees were unaware of the voice icon; only 18% had tried voice in the last month. Heatmaps show low interaction with the mic entry point on home and chat screens.
- Reliability issues: 27% reported mid-session dropouts; 31% cited misrecognitions on names, acronyms, and domain terms. Voice CSAT trails text by 0.6 points (3.7 vs 4.3/5).
- Workflow friction: Diary studies show users default to “Hey Google/Siri” for quick notes while commuting, then copy results into chat—adding 2–3 extra steps and exposing data to non-approved channels.
- Latency/continuity: P95 time to first token is 1.4s for voice vs 0.9s for text; long prompts (>45s) have a 9% higher interruption rate.

Anecdotes

- **SAURAV - JOB SEEKER** - “*I can’t find the mic unless I’m already in a chat—by then I’ve typed.*”
- **DEV - WORKING AS SOFTWARE DEVELOPER** - “Long technical prompts get clipped halfway, so I switch to my phone’s assistant.”
- **DISHA - JOB SEEKER** - “I don’t use it during interviews prep because it misunderstands company/product names.”



Competitive insights

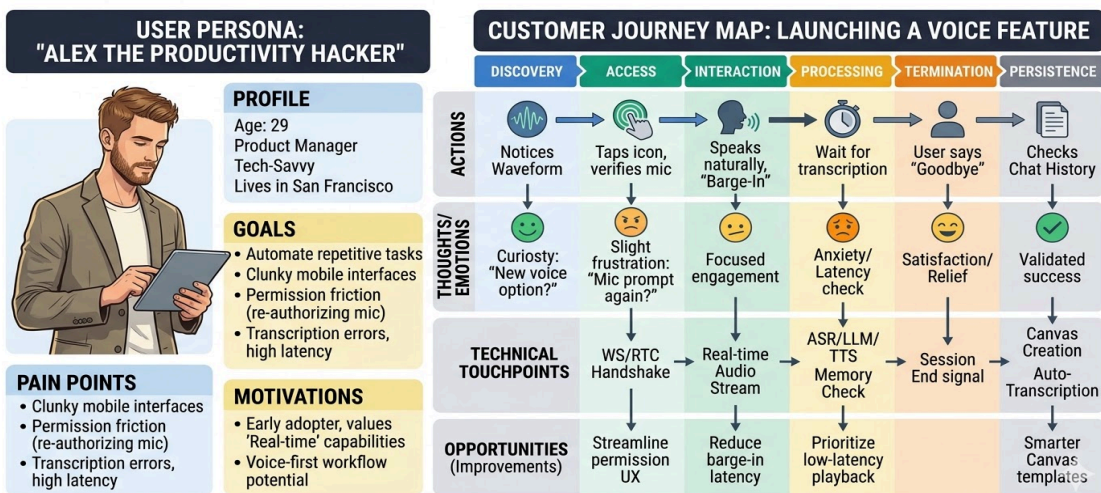
- Assistants with “**Hey chatgpt**” (Google Assistant, Siri, Alexa) show higher hands-free initiation and habit formation; their wake-word false accept rates are <1% with on-device detection.
- Leading productivity apps surface “press-to-speak” prominently and offer persistent push-to-talk to reduce accidental activation.
- Competitors market “enterprise-grade voice” with on-device wake-word, opt-in privacy controls, and domain-adapted vocabularies—reducing misrecognitions on proper nouns.

- Gap/opportunity: Few enterprise chat tools offer secure wake-word activation inside approved apps; differentiation is reliability (WER), domain vocab, and clear privacy controls.

Understanding the target audience

- **Segment/size:** Mobile/hybrid employees and interview-prep users; ~25–35% active users on mobile weekly, 10–15% frequent dictation.
- **Personas/journeys:**

USER PERSONA & JOURNEY



Unmet needs:

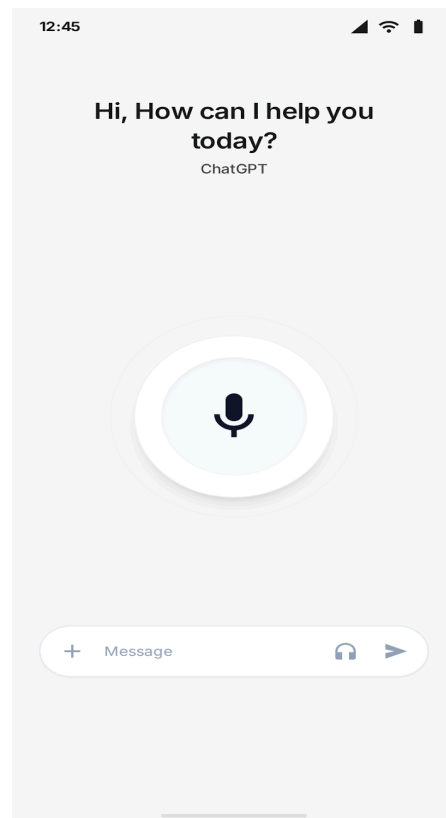
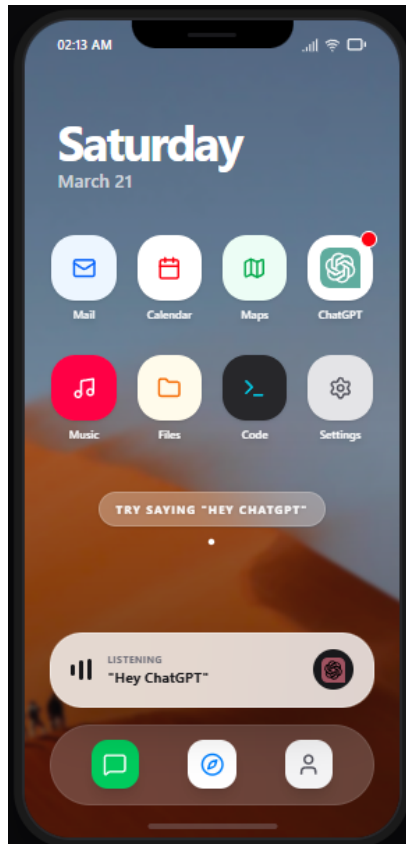
- Faster hands-free start; mic is hard to find.
- Reliability for long dictation; dropouts/clipping.
- Higher accuracy on acronyms/proper nouns.
- Lower latency and partial transcript feedback.
- Clear privacy/controls to stay within approved channels.

Solution overview

- Introduce a voice-first experience with optional wake word **"hey chatgpt"** and prominent push-to-talk, improved ASR tuned for domain terms, and resilient streaming to prevent dropouts. This is informed by user research on discoverability/reliability gaps, prior mic placement/latency experiments, and competitive benchmarks emphasizing on-device, opt-in wake-word privacy.

- Add a proactive, **dismissible greeting on app open** (“Hi there — how can I help?”) with clear voice and text entry points. The greeting respects user preference(enable/disable, snooze), quiet hours, and enterprise privacy settings; no audio is captured until explicit user action.

User flows/wireframes



Key features

- Dismissible greeting with clear controls (snooze/disable) to avoid interruption.
- Context-aware suggestions (e.g., “Continue last task,” “Draft interview questions”) without auto-sending any data.
- Privacy-first defaults: greeting is rendered locally; no audio is recorded until explicit user action.

Key logic and data changes (addition)

- Preference flags: greeting_enabled (bool), snooze_until (timestamp), last_dismiss_reason (enum).
- Context rules: show only when user is idle on open, not in active call, and within allowed hours; respect MDM/policy flags.
- Telemetry: greeting_impressions, greeting_action (start_voice/start_text/dismiss), downstream session start and completion; measure lift in activation vs control.

Notes on compliance and user experience

- No auto-recording or auto-sending; activation remains explicit to meet enterprise privacy expectations.
- Provide in-settings controls and first-run onboarding explaining the greeting and data handling.

Launch Readiness

- Key milestones: Design complete (Day 14); Dev complete (Day 45); QA/perf/accessibility sign-off (Day 60); Dogfooding (Days 61–75); Pilot rollout (Day 76); GA on green metrics (Day 90).
- Checklist: Support docs/training; SRE capacity/alerts; Security/Privacy/Legal reviews and disclosures; Analytics dashboards live; QA/device matrix; GTM comms; Accessibility checks; Feature flags and rollback plan.
- Experiments: A/B persistent mic vs floating; A/B proactive greeting; ASR domain-bias tuning; network resilience variants; staged rollout with holdouts.

Open Questions & Decisions Taken

- Open questions (with due dates as relative days): Wake word scope (Day 30); initial language coverage (Day 30); default retention window (Day 35); final discoverability pattern (Day 50); accessibility scope at launch (Day 55).
- Descoped: OS-level/global hotword; third-party assistant integrations; broad multilingual or offline ASR; advanced analytics/coaching.
- Trade-offs: Reliability over breadth; privacy-first opt-in wake word; minor UI clutter for discoverability; higher compute for lower latency; faster launch by deferring riskier features.